

NORAH HEAD LH, Australia

WMO: 957700

Lat: 33.2814S

Lon: 151.5767E

Elev: 64

StdP: 14.66

Time Zone: 10.00 (AUS)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	46.2	47.8	30.7	25.0	57.8	33.1	27.6	55.7	33.0	59.7	29.1	59.1	8.1	290	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.0	85.4	69.2	81.6	70.8	79.4	70.6	75.0	79.8	73.8	78.1	72.8	76.9	11.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.6	125.3	77.1	72.4	120.2	76.3	71.3	115.8	75.5	38.5	80.1	37.4	78.3	36.5	77.1	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.4	28.4	25.0	42.7	98.1	1.6	4.2	41.5	101.2	40.5	103.6	39.6	106.0	38.4	109.1		
(5)			38.5	77.3	1.7	1.6	37.2	78.5	36.2	79.4	35.2	80.3	33.9	81.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	72.7	73.0	71.4	66.8	61.8	57.9	56.3	58.0	61.8	65.2	67.7	70.5	(6)	(7)
	DBStd	6.93	3.52	3.34	3.24	3.75	3.76	3.29	3.16	4.04	4.44	4.83	4.27	3.82	(7)	(8)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	(9)
	HDD65	1032	0	0	1	22	112	214	270	221	116	54	18	4	(9)	(10)
	CDD50	5557	704	645	665	504	366	236	196	249	354	472	531	635	(10)	(11)
	CDD65	1114	239	226	201	76	13	1	0	4	20	61	99	174	(11)	(12)
	CDH74	3099	727	702	442	131	13	0	0	16	101	258	261	448	(12)	(13)
	CDH80	553	122	79	52	18	0	0	0	1	23	94	79	85	(13)	(14)
(14) Wind	WSAvg	11.5	13.3	13.0	12.4	11.1	9.8	9.9	9.0	10.1	11.3	12.3	13.2	13.2	(14)	(15)
(15) Precipitation	PrecAvg	47.5	4.2	5.6	5.1	4.6	5.2	4.9	2.8	2.8	3.0	2.9	3.8	3.0	(15)	(16)
	PrecMax	72.0	17.3	23.9	13.3	16.4	17.5	16.7	6.3	13.1	11.4	8.0	9.1	9.3	(16)	(17)
	PrecMin	28.5	0.3	0.3	0.5	0.2	0.2	0.1	0.0	0.1	0.0	0.0	0.5	0.3	(17)	(18)
	PrecStd	10.7	3.4	5.2	3.3	4.2	3.8	3.8	1.9	2.7	2.8	2.2	2.1	2.2	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.6	86.4	85.1	82.1	75.8	70.5	71.1	76.2	83.5	89.8	89.9	90.0	(19)	(20)
		MCWB	71.4	75.0	73.1	67.9	63.7	60.5	58.1	59.4	62.8	66.0	67.0	68.4	(20)	(21)
	2%	DB	82.4	81.8	80.3	77.3	72.3	67.2	66.7	70.6	76.5	81.0	79.7	80.9	(21)	(22)
		MCWB	73.7	73.4	71.9	67.6	63.2	59.4	56.4	57.5	60.8	63.4	67.6	70.1	(22)	(23)
	5%	DB	79.6	79.7	77.9	74.7	69.9	65.2	64.3	67.4	71.7	74.9	75.9	78.1	(23)	(24)
		MCWB	72.4	72.5	70.7	66.7	62.3	58.5	55.7	57.1	60.4	64.4	67.6	69.8	(24)	(25)
	10%	DB	77.6	77.9	76.2	72.7	68.1	63.7	62.4	65.1	68.8	71.7	73.4	76.0	(25)	(26)
		MCWB	71.1	71.5	69.8	65.8	61.7	57.7	55.2	56.4	59.9	63.7	66.7	68.8	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.5	77.0	75.5	72.0	68.0	64.4	62.2	63.7	66.6	70.1	72.2	74.2	(27)	(28)
		MCDB	82.4	82.1	79.6	76.5	70.8	67.4	65.7	69.7	73.1	76.9	79.4	79.7	(28)	(29)
	2%	WB	74.8	75.1	73.7	69.9	66.2	61.7	59.7	61.3	64.6	68.0	70.3	72.6	(29)	(30)
		MCDB	79.8	79.6	77.9	74.1	69.5	65.3	63.6	66.4	70.2	73.6	75.8	77.5	(30)	(31)
	5%	WB	73.6	73.8	72.5	68.5	64.5	60.3	58.0	59.7	63.3	66.5	69.0	71.3	(31)	(32)
		MCDB	77.8	77.9	76.5	72.8	68.5	63.8	62.1	64.8	68.4	71.5	73.9	75.8	(32)	(33)
	10%	WB	72.4	72.8	71.3	67.2	62.8	59.0	56.5	58.2	62.0	65.3	67.8	70.1	(33)	(34)
		MCDB	76.4	76.7	75.0	71.7	67.1	62.9	60.9	63.4	67.0	69.9	72.5	74.1	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.0	9.3	10.8	11.2	10.7	12.0	13.2	12.9	12.2	10.5	9.8	(35)	(36)
		MCDBR	13.3	12.2	12.2	13.8	13.7	12.3	14.6	16.6	18.3	18.7	15.8	14.0	(36)	(37)
	5% WB	MCWBR	6.7	6.5	7.1	8.5	8.5	8.4	9.3	10.0	10.1	9.9	8.0	7.4	(37)	(38)
		MCWBR	12.0	10.6	11.3	11.8	11.1	10.5	11.6	13.7	14.5	14.6	12.9	11.9	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.398	0.389	0.359	0.334	0.318	0.311	0.302	0.311	0.344	0.368	0.392	0.388	(39)	(40)
		taud	2.451	2.498	2.575	2.617	2.612	2.626	2.636	2.599	2.501	2.462	2.435	2.466	(40)	(41)
	Ebn at Noon		298	294	293	283	272	266	275	287	292	298	298	302	(41)	(42)
		Edh at Noon	38	35	30	26	23	22	22	26	32	36	38	38	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		2063	1813	1533	1228	962	767	897	1187	1564	1804	1939	2096	(43)	(44)
	RadStd		139	149	131	101	72	67	73	69	80	144	175	163	(44)	(45)

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (33 neighbors)	+0.81	N/A	N/A	+1.53	N/A	-4	-153	+340	+184	(47)

Nomenclature: See separate page